

AWS DevOps

Week 1



AWS Infrastructure

- Performance Excellence
- Security
- Availability
- Global Footprint
- Scalability
- Flexibility

arn:aws:iam::761xxxx63xx7:user/adminx
arn:aws-cn:iam::761xxxx63xx7:user/adminx



AWS Infrastructure

AWS Partitions

- GovCloud
- China Region Cloud
- Open Regions (with some restrictions)

▼ GovCloud (US)

[Sign up for AWS GovCloud \(US\)](#)



▼ Close Account

☐ I understand that by clicking this checkbox, I am closing my AWS account. The closure of my AWS account serves as notice to AWS that I wish to terminate the AWS Customer Agreement or any other agreement with AWS that governs my AWS account, solely with respect to that AWS account.

Monthly usage of certain AWS services is calculated and billed at the beginning of the following month. If I have used these types of services this month, then at the beginning of next month I will receive a bill for usage that occurred prior to termination of my account. In addition, if I have any active subscriptions (such as a Reserved Instance for which I have elected to pay in monthly installments), then even after my account is closed I may continue to be billed for the subscription until the subscription expires or is sold in accordance with the terms governing the subscription.

I acknowledge that I may reopen my AWS account only within 90 days of my account closure (the "Post-Closure Period"). If I reopen my account during the Post-Closure Period, I may be charged for any AWS services that were not terminated before I closed my account. If I reopen my AWS account, I agree that the same terms will govern my access to and use of AWS services through my reopened AWS account.

If I choose not to reopen my account after the Post-Closure Period, any content remaining in my AWS account will be deleted. For more information, please see the

☐ I understand that after the Post-Closure Period I will no longer be able to reopen my closed account.

☐ I understand that after the Post-Closure Period I will no longer be able to access the Billing Console to download past bills and tax invoices.

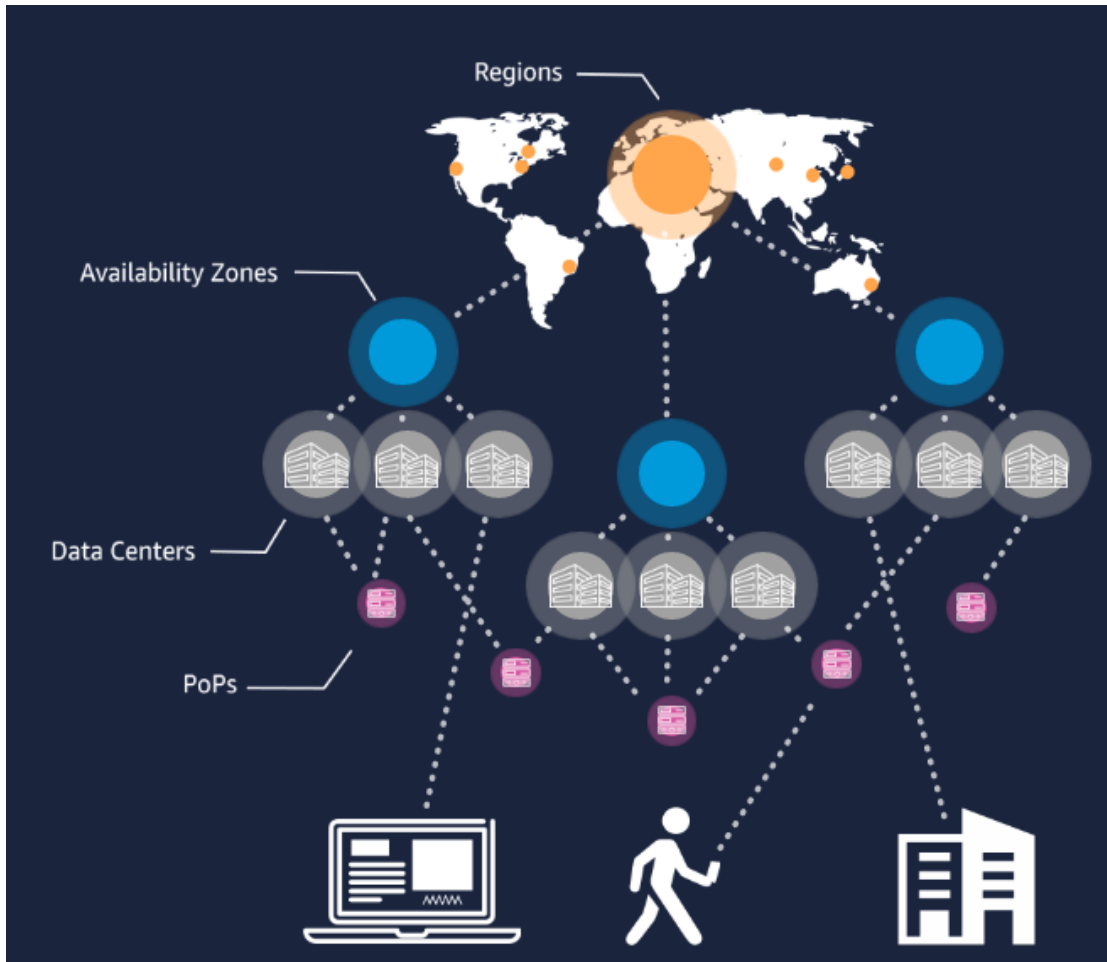
If you wish to [download any statements you can do so here](#). Select the month and expand the summary section to download the payment invoices and/or tax documents.

☐ I understand that after the Post-Closure Period I will not be able to create a new AWS account with the email address currently associated with this account.

If you wish to update your e-mail address, [follow the directions here](#).

[Close Account](#)

AWS Service Status



<https://status.aws.amazon.com/>

Amazon Route 53 Private DNS (N. California)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (N. Virginia)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (Ohio)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (Oregon)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Resolver (N. Virginia)	✓	✓	✓	✓	✓	✓	✓	
[RESOLVED] Increased Email Receiving Latencies 1:46 PM PDT We are currently investigating elevated latencies in our email-sending APIs in the US-EAST-1 Region. This includes the SendEmail/SendRawEmail APIs as well as calls made to the SMTP endpoint. 2:20 PM PDT Between 12:25 PM and 1:52 PM PDT we experienced elevated delivery delays in the US-EAST-1 Region for mail sent to a specific external email provider. This issue impacted the SendEmail/SendRawEmail APIs as well as calls made to the SMTP endpoint. All delayed emails have now been delivered. The issue has been resolved and the service is operating normally.								
Amazon Simple Email Service (Oregon)	✓	✓	✓	✓	✓	✓	✓	
SNS (Montreal)	✓	✓	✓	✓	✓	✓	✓	

Personal Health
Dashboard

Dashboard

Event log

Integrations

Organizational View API

Set up CloudWatch

Events 

Dashboard

2 Open issues
Past 7 days

0 Scheduled changes

0 Other notification
Past 7 days

Issues that might affect your AWS infrastructure. [0 issues](#) were resolved in the past 24 hours. [See](#)



Start time: June 5, 2020 at 4:53:38 PM UTC+5:30 [Add filter](#)

	Event	Region... 	Start time	Last update time	Affected
☒	IAM operational issue	-	June 12, 2020 at 12:...	June 12, 2020 at 4:0...	-
☐	Organizations operatio...	-	June 12, 2020 at 1:2...	June 12, 2020 at 2:5...	-

multiple regions. IAM role creation is impacted. Other AWS services like AWS CloudFormation whose features require these actions may also be impacted. User authentications and authorizations are not impacted.

[02:12 AM PDT] We have identified the root cause of the increased error rates and latencies on the AWS IAM CreateRole and CreateServiceLinkedRole APIs and are working towards resolution. Other AWS services such as AWS CloudFormation whose features require these actions may also be impacted. User authentications and authorizations are not impacted.

[03:30 AM PDT] We wanted to provide you with more details on the issue causing increased error rates and latencies on the AWS IAM CreateRole and CreateServiceLinkedRole APIs. While we have identified the root cause and are working towards resolution, with an issue like this, it is always difficult to provide an accurate ETA, but we expect to restore access to the CreateRole and CreateServiceLinkedRole APIs within the next several hours. We are working through the recovery process now and will continue to keep you updated if this ETA changes. IAM user authentications and authorizations are not impacted. Other AWS services like AWS CloudFormation whose features require these actions may also be impacted.

Event data

Event	IAM operational issue
Status	Open
Region/AZ	-
Start time	June 12, 2020 at 12:00:13 PM UTC+...
End time	-
Event category	 Issue



Personal Health Dashboard

Dashboard

Event log

Integrations

Organizational View API

Set up CloudWatch

Events

Dashboard

2 Open issues
Past 7 days

0 Scheduled changes

0 Other notification
Past 7 days

Issues that might affect your AWS infrastructure. 0 issues were resolved in the past 24 hours.

See

Start time: June 5, 2020 at 4:48:38 PM UTC+5:30 Add filter					
	Event	Region... i	Start time	Last update time	Affected resources
☐	IAM operational issue	-	June 12, 2020 at 12:...	June 12, 2020 at 4:0...	-
☒	Organizations operatio...	-	June 12, 2020 at 1:2...	June 12, 2020 at 2:5...	-

Organizations operational iss... Close



Details

Affected resources

Increased API Error Rates

[12:57 AM PDT] We are investigating increased error rates for account creation and invitations for Organizations with all features enabled. Other AWS services using Organizations service trust may also be impacted when activating new organizations or accounts.

[02:13 AM PDT] We have identified the root cause of the increased error rates for account creation and invitations for Organizations with all features enabled and are working towards resolution. Other AWS services using Organizations service trust may also be impacted when activating new organizations or accounts.

Event data

Event	Organizations operational issue
Status	Open
Region/AZ	-
Start time	June 12, 2020 at 1:27:50 PM UTC+5:30
End time	-
Event category	Issue



Services ▾

Resource Groups ▾



Global ▾

Support ▾

Personal Health
Dashboard

Dashboard

Event log

Integrations

Organizational View API

Set up CloudWatch

Events

Event log

Event category: Issue Add filter

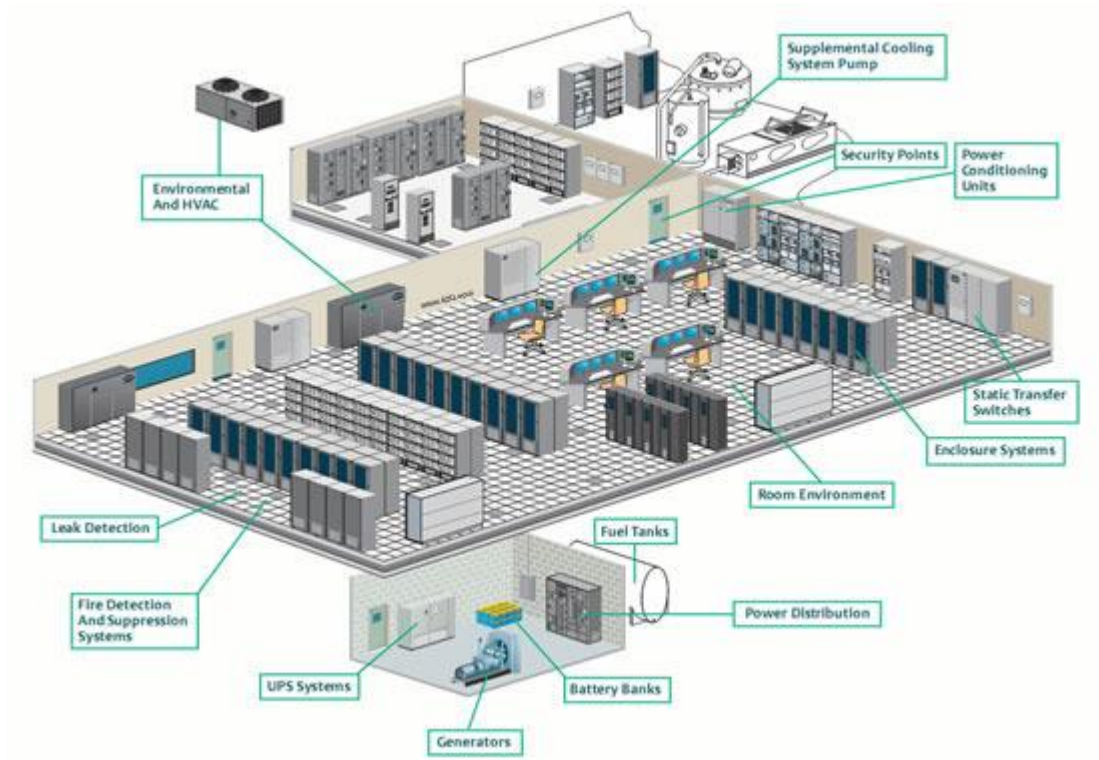
	Event	Status	Region/AZ	Start time	Last update time	Affected resources	Event category
☐	IAM operational issue	Open	-	June 12, 2020 at 12:00:1...	June 12, 2020 at 4:57:30 ...	-	Issue
☐	Organizations operational is...	Open	-	June 12, 2020 at 1:27:50 ...	June 12, 2020 at 2:57:31 ...	-	Issue
☐	SNS operational issue	Closed	us-east-1	June 11, 2020 at 2:22:13 ...	June 11, 2020 at 3:17:51 ...	-	Issue
☐	DynamoDB operational issue	Closed	sa-east-1	June 6, 2020 at 6:46:23 A...	June 6, 2020 at 10:29:47 ...	-	Issue
☐	onal issue	Closed	us-west-2	June 4, 2020 at 3:50:48 A...	June 4, 2020 at 6:23:01 A...	-	Issue
☐	rRegistry op...	Closed	us-west-2	May 23, 2020 at 4:18:49 ...	May 23, 2020 at 4:19:31 ...	-	Issue
☐	ervice operati...	Closed	us-east-1	May 21, 2020 at 8:14:30 ...	May 21, 2020 at 9:08:20 ...	-	Issue
☐	al issue	Closed	us-west-1	May 10, 2020 at 8:08:49 ...	May 10, 2020 at 10:25:46 ...	-	Issue
☐	rService ope...	Closed	us-west-1	May 10, 2020 at 8:01:30 ...	May 10, 2020 at 10:22:25 ...	-	Issue
☐	operational ...	Closed	us-west-2	May 1, 2020 at 2:17:10 A...	May 1, 2020 at 3:41:31 A...	-	Issue
☐	rational issue	Closed	-	April 22, 2020 at 4:27:30 ...	April 22, 2020 at 4:58:15 ...	-	Issue
☐	l issue	Closed	us-east-1	April 21, 2020 at 3:15:25 ...	April 21, 2020 at 3:46:44 ...	-	Issue
☐	SQS operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:12:27 ...	April 20, 2020 at 9:58:00 ...	-	Issue
☐	Cloudwatch operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:08:47 ...	April 20, 2020 at 8:46:13 ...	-	Issue
☐	Lambda operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:15:34 ...	April 20, 2020 at 8:44:49 ...	-	Issue
☐	CloudFormation operational ...	Closed	ap-northeast-1	April 20, 2020 at 4:39:46 ...	April 20, 2020 at 8:43:38 ...	-	Issue

0 Other notifications
Past 7 days

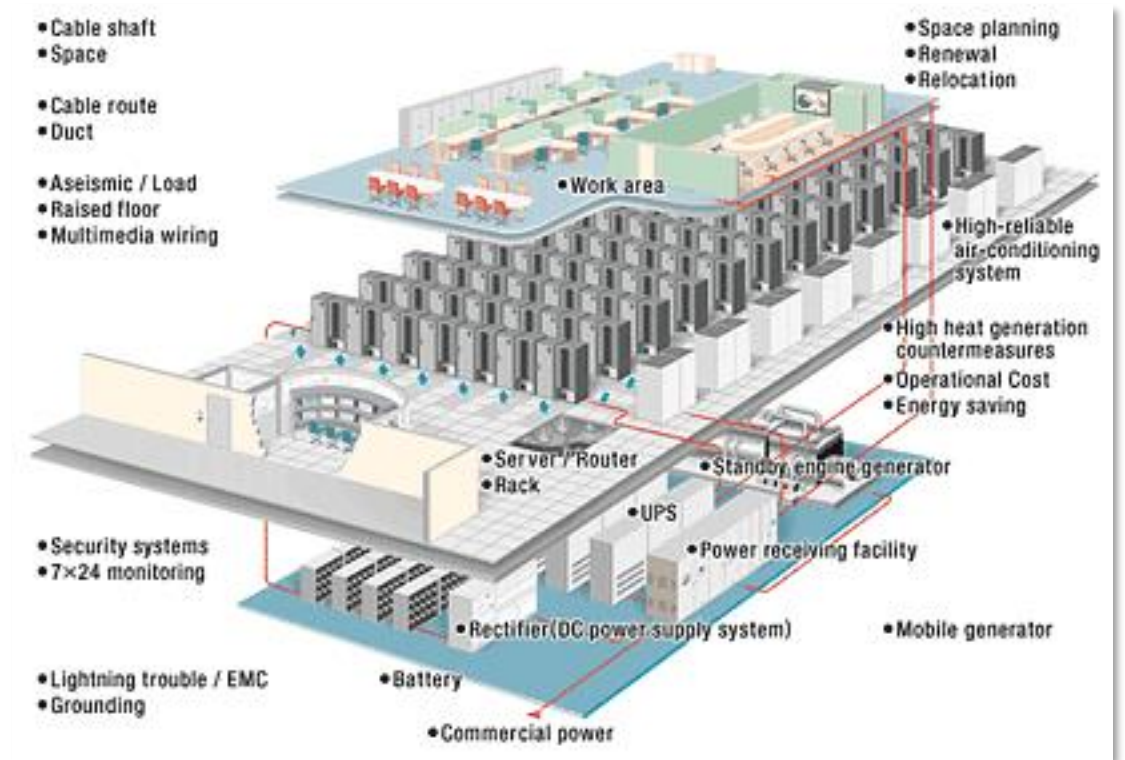
[See all issues](#)



Data centre Architecture and Infrastructure



<https://www.mainlinecomputer.com/t/product-lines/infrastructure/airflow-management-solutions/data-center-assessment/>



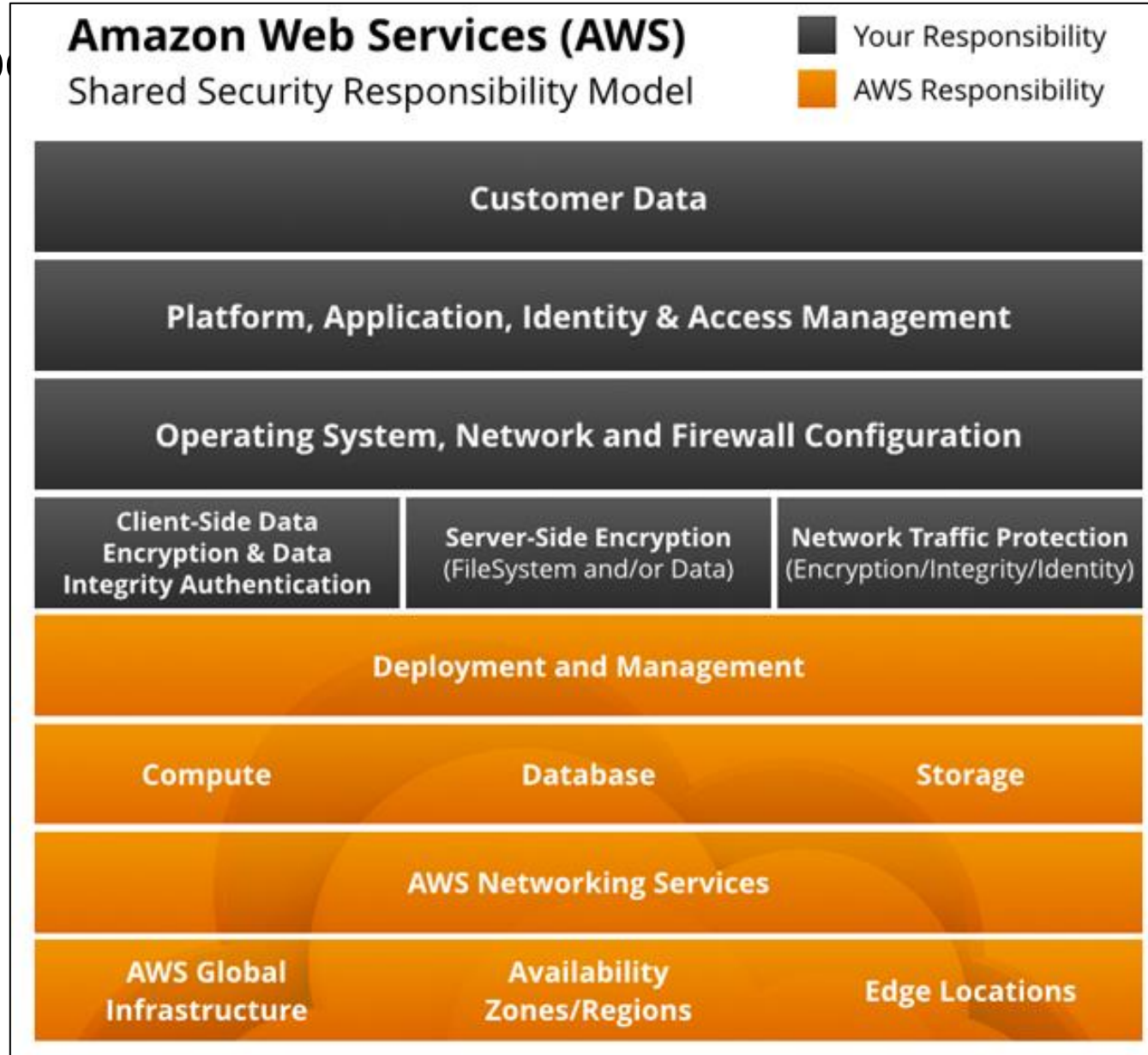
<https://www.pinterest.com/pin/549861435748646461/>

Find out more on Tower Servers, Rack Servers and Blade Servers

Identity and Access Management (IAM) – Best Practice

- Use Groups wherever possible
- Grant least privilege
- Define and implement a strong password policy
 - Standard + past 4 password cannot be repeated
- For cross-account access always use Roles
- Use Explicit deny actions for added security
- Never Share credentials
- Never Email, print or publish to repos
 - User name, passwords, Access keys and Secret keys
- Use Multiple accounts for isolation
- Important:
 - Never use root account for day-to-day activities
 - Enable MFA for root account – best practice

Shared Responsibility



1. How to access AWS

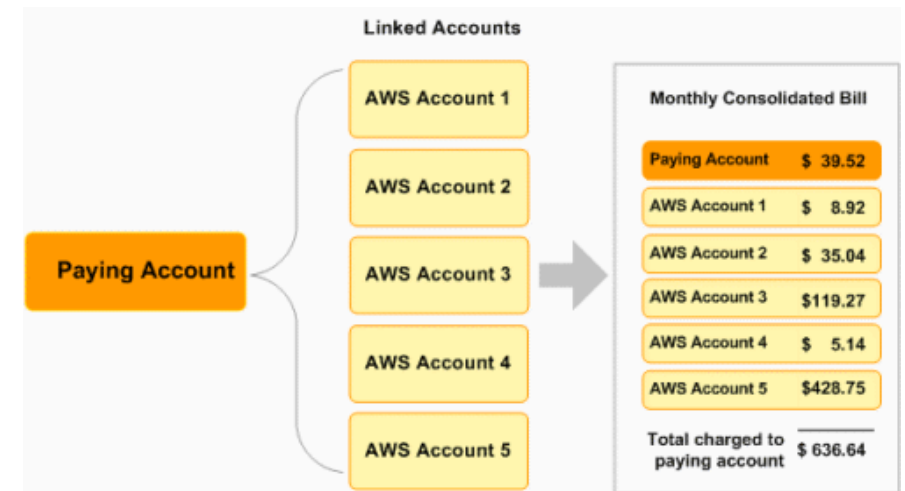
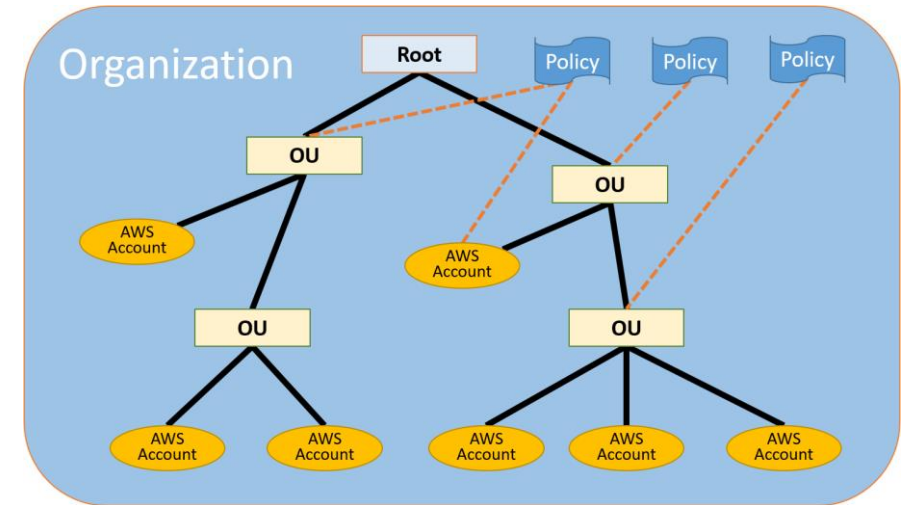
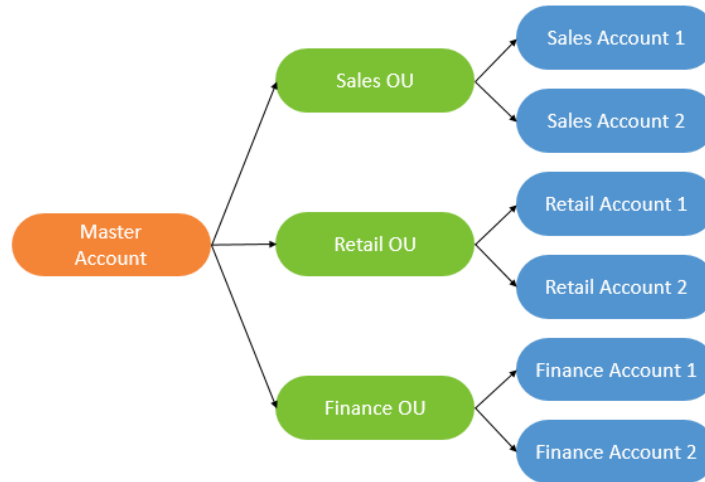
- Create an individual account with AWS
 - by default
 - You get a general account within AWS partition
 - govcloud no access by default &
 - Partition aws-cn (China Beijing regions) only for accounts created within China
 - All regions are available for use
 - *with few regional restrictions*
 - One account can have resources in multiple regions
 - Cost for similar resources may vary from region to region
 - Account creation requires valid credit-card details
 - New account enjoy 12-months free usage, see details [here](https://aws.amazon.com/free) (aws.amazon.com/free)

2. Organization Account

AWS Organization

- Organization account

- Can have multiple account, one master account
- Hierarchy
 - Multiple accounts under master account
 - Centrally manage all accounts and related policies
 - Volume discounts for EC2 and S3
 - Consolidated billing facility & much more

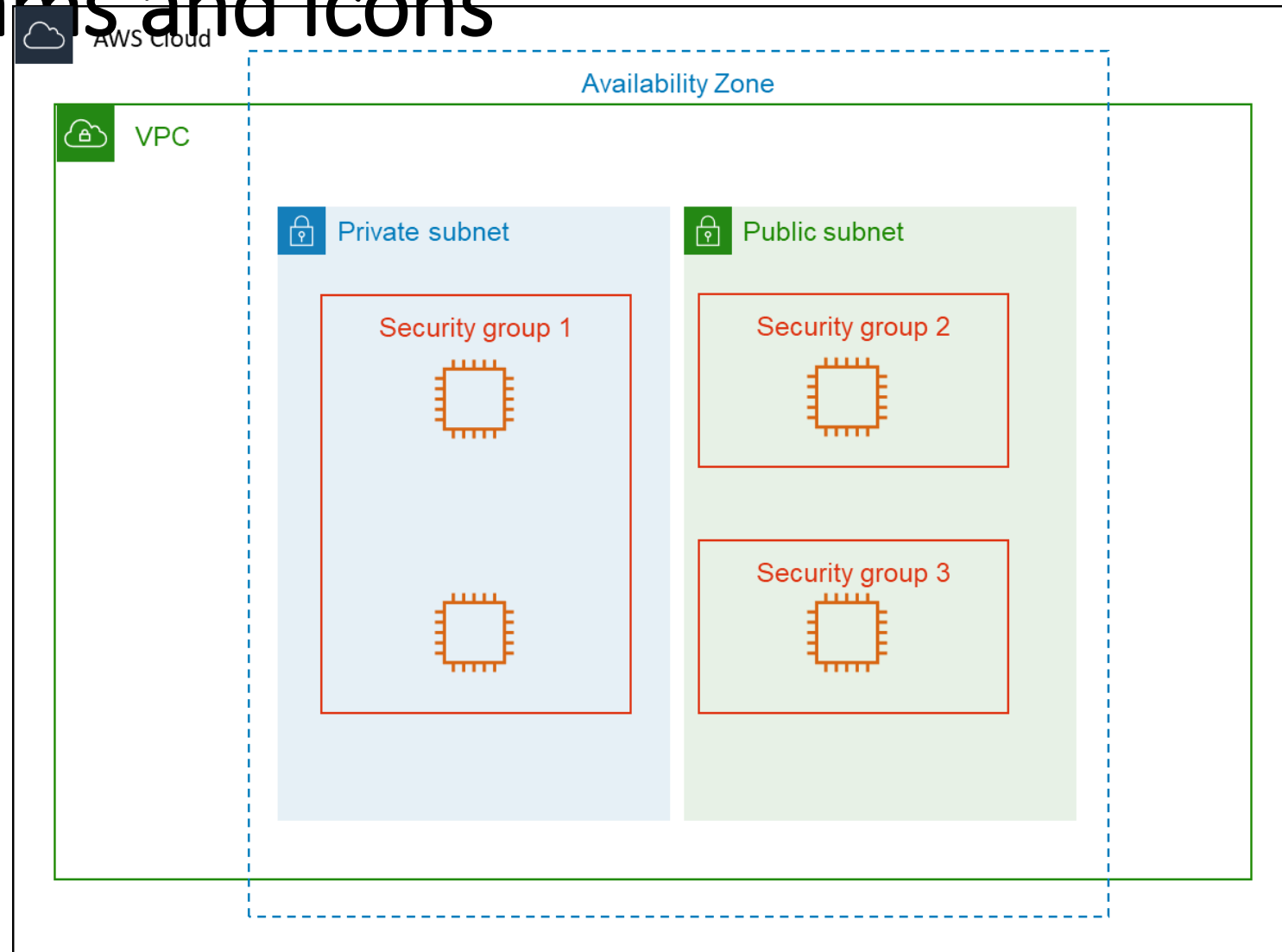


For simplicity purpose we will use individual account for hands-on work

Architecture Diagrams and Icons

- Use these icons for consistent and standard way to document your cloud architecture
- [Icons](#) available for
 - PowerPoint

[Reference Architecture solution Samples](#)



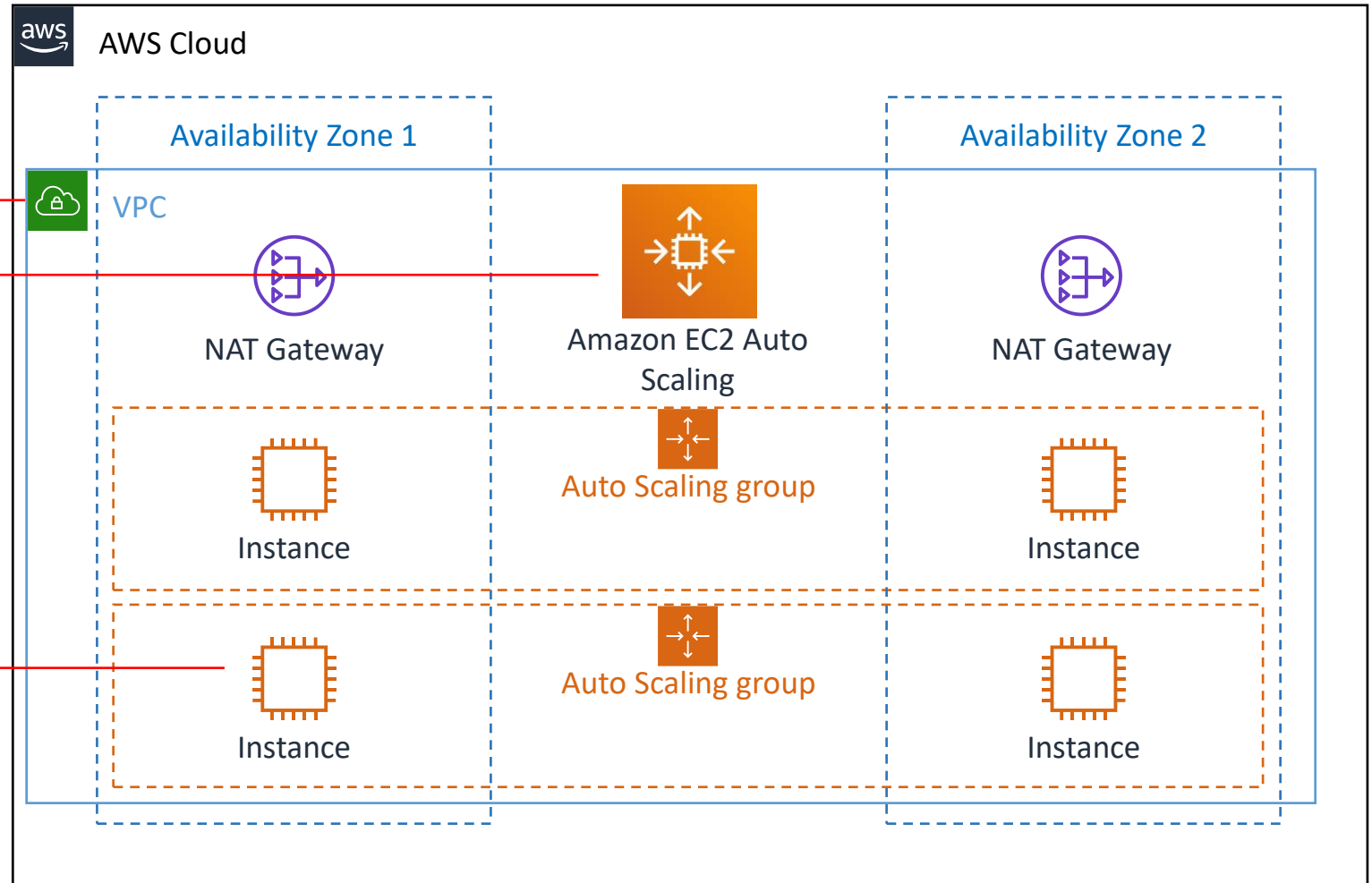
Architecture Diagrams and Icons

Group icon

Layered groups

Product icon

Resource icon



VPC Next Topics

- Subnets
- Internet Gateway
- NACL

Subnets

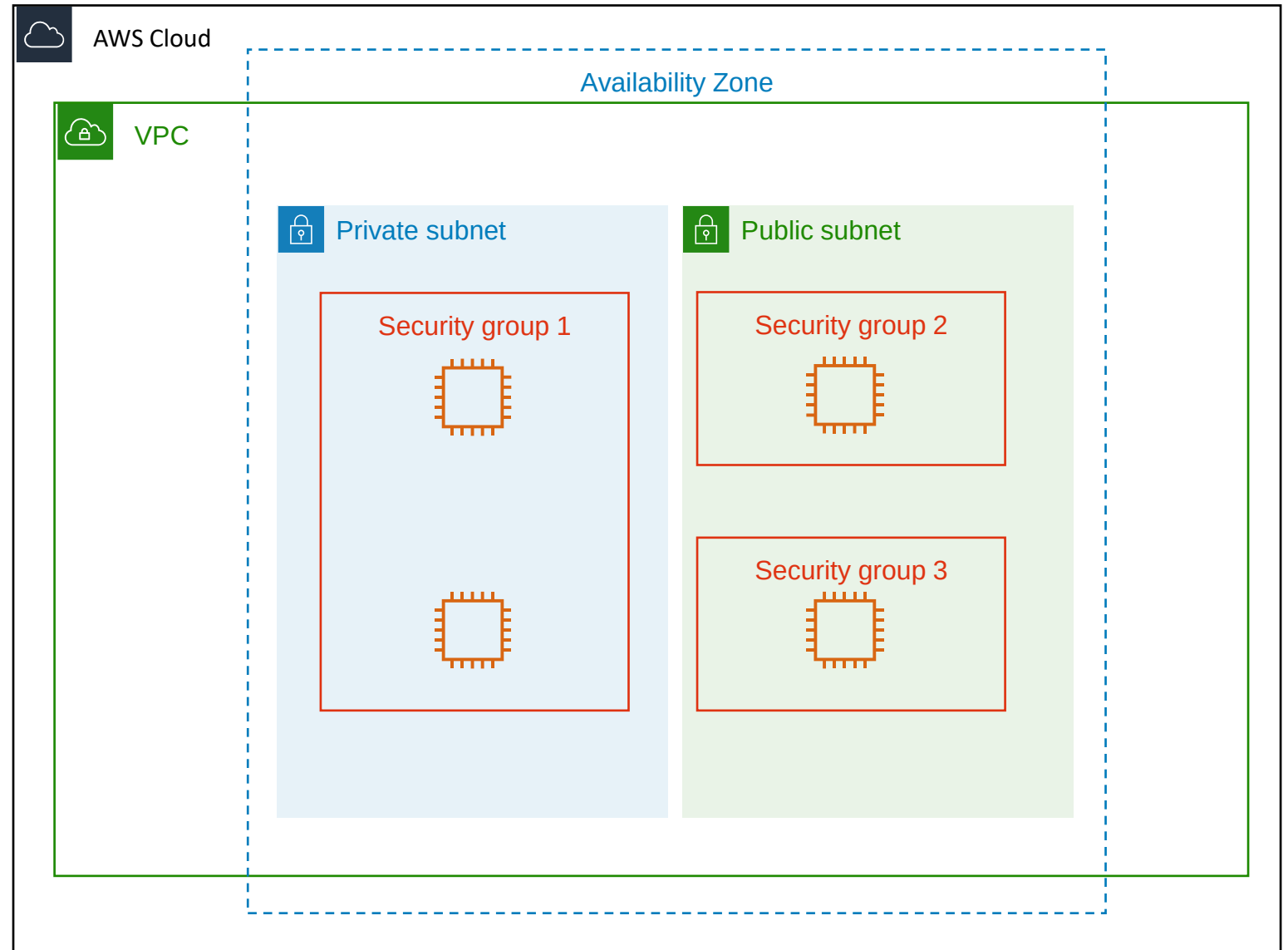
- Private
- Public
- Route table

IAM

- User
- Groups
- Policies
- Role

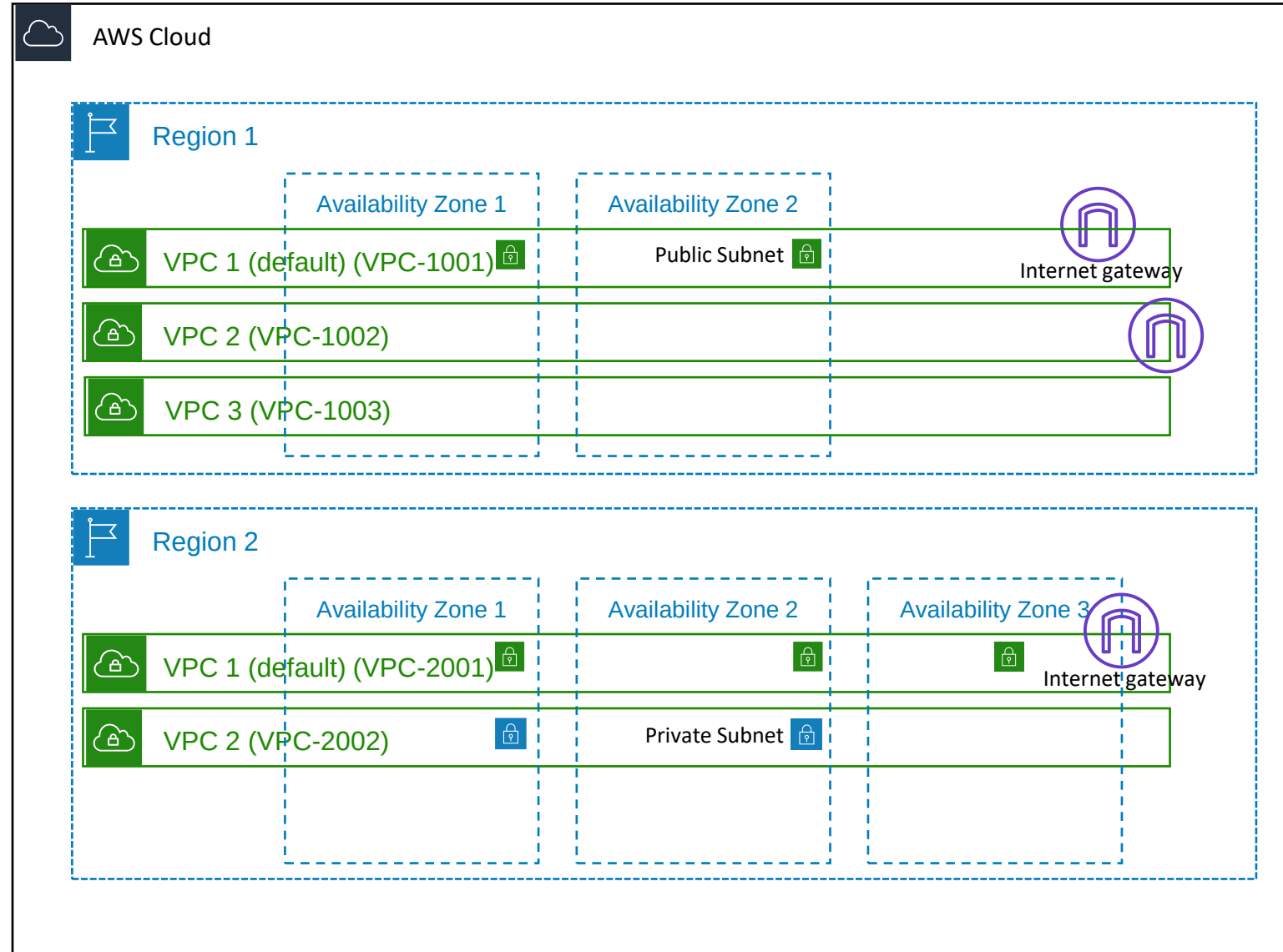
Launching an EC2

- Instance Types
- Pricing Models
- Vertical Scaling
- Elastic IP
- Security Group



• VPC, Subnets & IP address

- Specific to a region
- One default VPC
 - One default subnet per AZ
 - One Internet gateway
 - Route table defined
 - Can have multiple VPCs
- Max five VPCs allowed
 - Limit can be increased via AWS support ticket
- [CIDR](#) block example
 - 172.31.0.0/16
 - 172.31.10.0/24
 - 4 addresses are reserved for internal use
 - First three
 - Last one for broadcasting
- [IP Address - Wiki](#)
- [NACL](#)
 - Network Access Control List
 - Default allowed
 - Explicit Deny



- # Identity and Access Management (IAM)

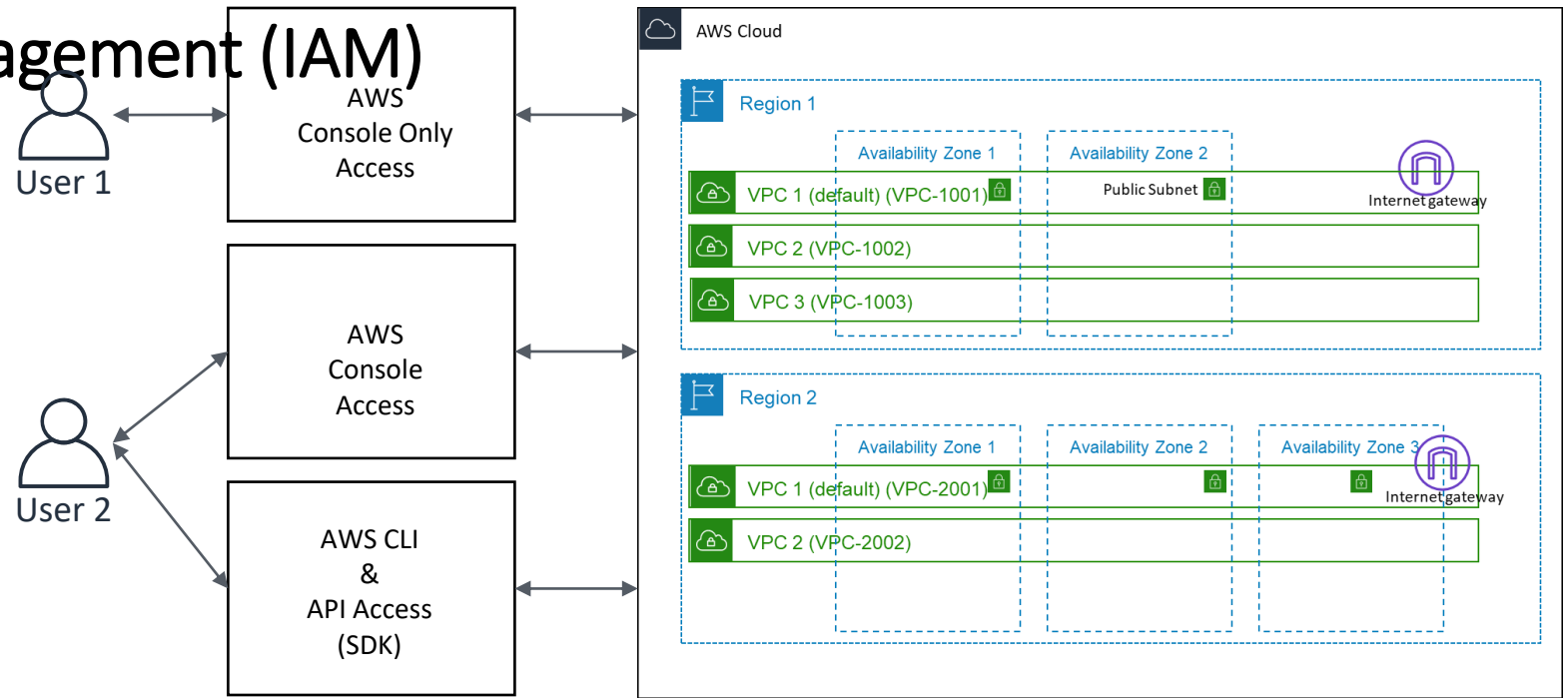
- Access Modes
 - Console Only
 - Console + Programmatic

- IAM - [Identity and Access Management](#)

- Authentication
- Authorization
- Users
- Groups
- Password Policy
- Multifactor Authentication

- [Billing Access](#)

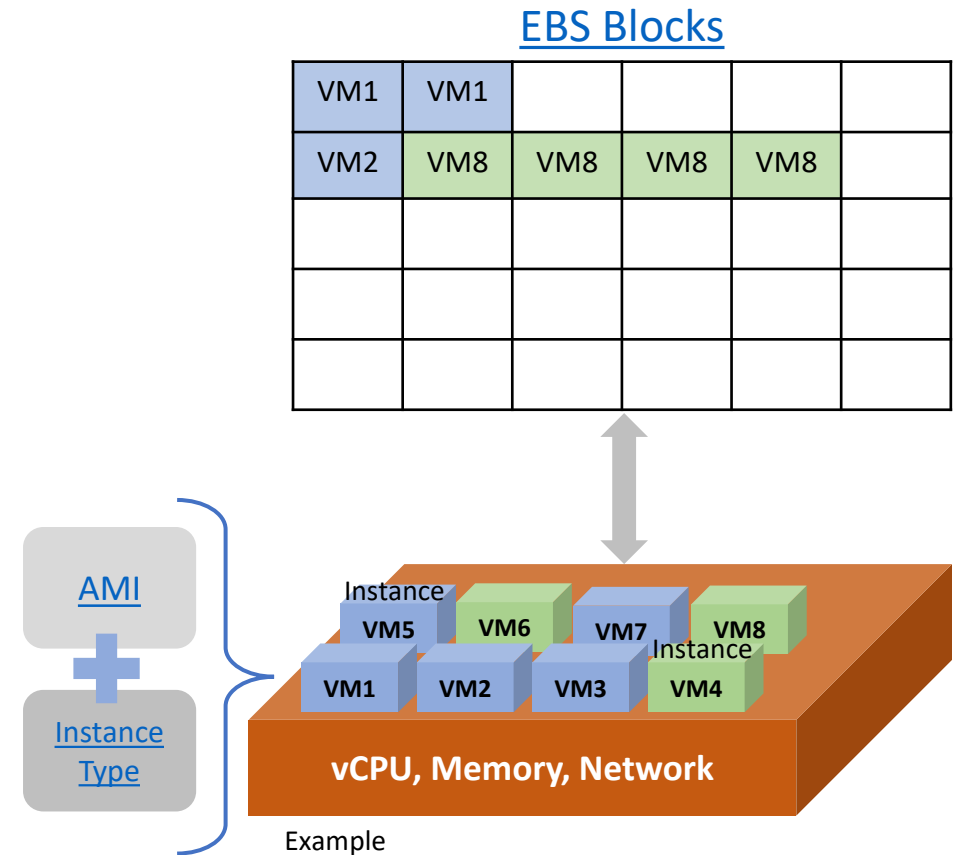
- Tags and Usage (ABAC)



```
Sample conditional Policy using ABAC
{
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:*"
      ],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalTag/S3Allowed": "Yes"
        }
      }
    }
  ]
}
```


EC2 – Elastic Cloud Compute

- **AMI**
 - Amazon Machine Image. Preconfigured templates of an OS + additional software
- **Instance Type**
 - Various configuration for of CPU, memory, storage and networking capacity
- **Instance**
 - VM configured for computing purpose using AMI + Instance Type
 - [Instance State](#)
- **EBS**
 - Elastic Block Storage, persistent storage volume for storing data
- **Instance Store**
 - Temporary data store, deleted when an instance is stopped or terminated
- **IPv4 address**
 - Private, Public and Elastic
- **[Security Group](#)**
 - Instance firewall, network defence layer along with subnet level NACL
- **[User Data](#)**
 - Set of commands executed on an EC2 instance when it is launched



Example

[Intel® Xeon® Platinum 9282 Processor](#)

56 cores, 112 threads, 2.6 GHz to 3.8 GHz

One physical hardware can run 1000x of VMs based on available vCPU, memory & [network](#)

One data centre has 1000x of physical hardware machine running [virtualization](#) software

Virtualization and AWS

- **PV** → Paravirtual
- **HVM** → Hardware Virtual Machine.
 - Is a type of instance that mimics bare-metal server setup which provides better hardware isolation. With this instance type, the OS can run directly on top of the Virtual Machine without additional configuration making it to look like it is running on a real physical server
- Linux Amazon Machine Images use one of two types of virtualization: paravirtual (PV) or hardware virtual machine (**HVM**). The main differences between PV and **HVM** AMIs are the way in which they boot and whether they can take advantage of special hardware extensions (CPU, network, and storage) for better performance.
- What does **hypervisor** do → The latest AWS hypervisor, **Nitro**, uses everything to provide a new hardware-assisted hypervisor that is easy to use and has near bare-metal performance. It's an exciting development in cloud computing: hardware virtualization is now fast
- **Host Machine** → Amazon EC2 relies on Xen Virtualization for launching all of its instances. Every physical machine has a hypervisor running on it. A Xen hypervisor allows multiple instances to share a single hardware platform. A hardware on which a hypervisor is running one or more virtual machines is called a Host machine

Working with AWS Storage (Available only for EC2)

- Attaching EBS volume as additional storage to an instance
 - Instance specific hence fixed to AZ of an instance
- EBS Backup Facility
- EFS – Elastic File Storage (try this additional exercise, optional)
 - Region Specific, available from all AZs within a region
- FSx – Windows File Server
 - Region Specific, available from all AZs within a region

For Linux

Follow these steps after attaching a new EBS volume to a Linux instance

- Mounting EBS to Linux
 - lsblk
 - Format this new volume
 - `sudo mkfs -t ext4 /dev/xvdf`
 - Create mount point
 - `sudo mkdir -p /mnt/data`
 - Mount this device to this mount point
 - `sudo mount /dev/xvdf /mnt/data`
 - df

Demo attaching EBS as external drive for Windows Instance